EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - II

Problem set: 2.1 Date: January 23, 2025

ADC Interfacing

Aim: To get familiarized with the on-chip ADC in the TM4C123GH6PM and convert an external analog signal into digital signal.

Components to be used:

- TIVA C Board
- Potentiometer

ADC on TM4C123GH6PM

- The TM4C123GH6PM microcontroller contains two identical ADC modules.
- These two modules, ADC0 and ADC1, share the same 12 analog input channels.
- Each ADC module operates independently and can therefore execute different sample sequences, sample any of the analog input channels at any time, and generate different interrupts and triggers.
- **Sample Sequencer:** The sampling control and data capture is handled by the sample sequencers. All of the sequencers are identical in implementation except for the number of samples that can be captured and the depth of the FIFO.
- **Trigger Source:** ADC can be triggered by various sources like processor, external GPIO, PWM, internal comparators, timer etc.
- **Relative Priority:** As mentioned earlier, there are 4 sample sequencers. Hence, we need to specify the relative priority of a sequencer in case we are using multiple sample sequencers.

Problem Definition: Given a 1-KΩ potentiometer, connect it to PE3 port of the TIVA C Board as given in the diagram below. Swipe the voltage between 0 to 3.3V and convert in to digital signal using on-chip ADC and display it on the CCS console.

General Instructions

- Include the following header filles in your code:
  "stdint.h", "stdbool.h", "inc/hw memmap.h", "inc/hw types.h", "driverlib/gpio.h", "driverlib/sysctl.h", "inc/hw ints.h", "driverlib/interrupt.h", driverlib/adc.h".
- Enable the system clock, ADC0 peripheral clock and the required port pins.
- Check the TM4C123GH6PM datasheet, Page 801 to find which pin can be configured as ADC input. Use GPIOPinTypeADC(), Page 266 to set.
- Set the enabled port pins to function as output pins.
- Configure the trigger source for ADC and priority of a sample sequence using the function ADCSequenceConfigure(), Page 38.
- Configure the ADC channel using the function ADCSequenceStepConfigure(),Page 42.
- Enable the sample sequence using the function ADCSequenceEnable(), Page41.

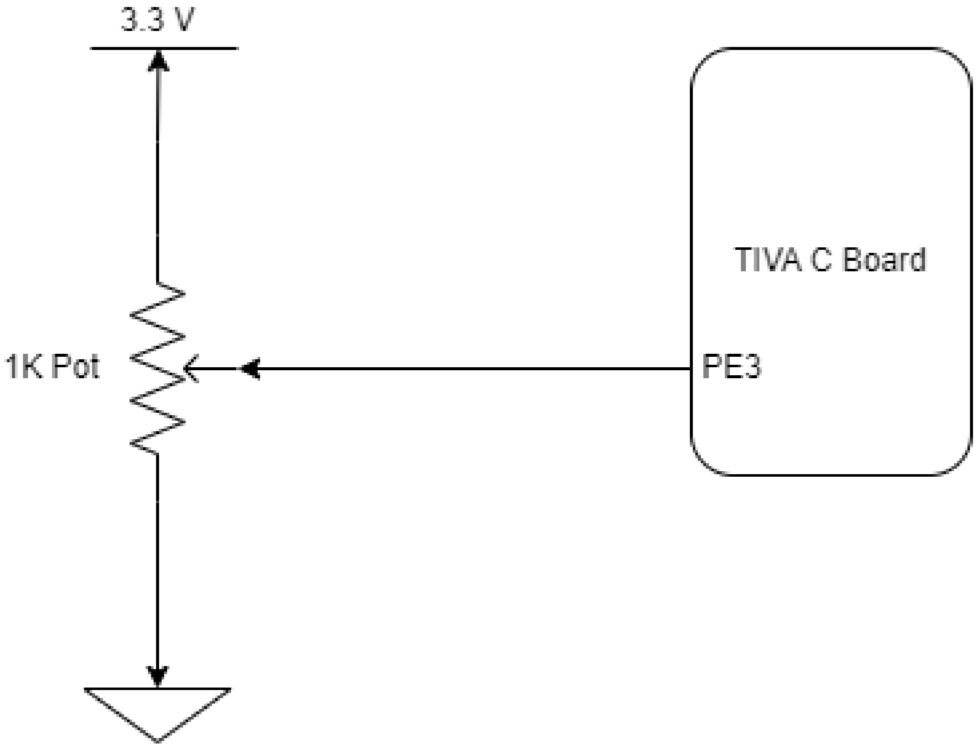

Figure 1: Potentiometer connection to TIVA C board

Console configuration:
Go through **Page 5-13 to 5-15** to configure the variable to display it in the CCS consle of the following link.
TM4C123G_LaunchPad_Workshop_Workbook

References:

1. TM4C123GH6PM : TM4C123GH6PM_Datasheet
2. TIVA Driver Library: TIVA_Driver_Library
3. TM4C123G LaunchPad Workshop Workbook TM4C123G_LaunchPad_Workshop_Workbook